Introduction to BSc Computer Science Degree

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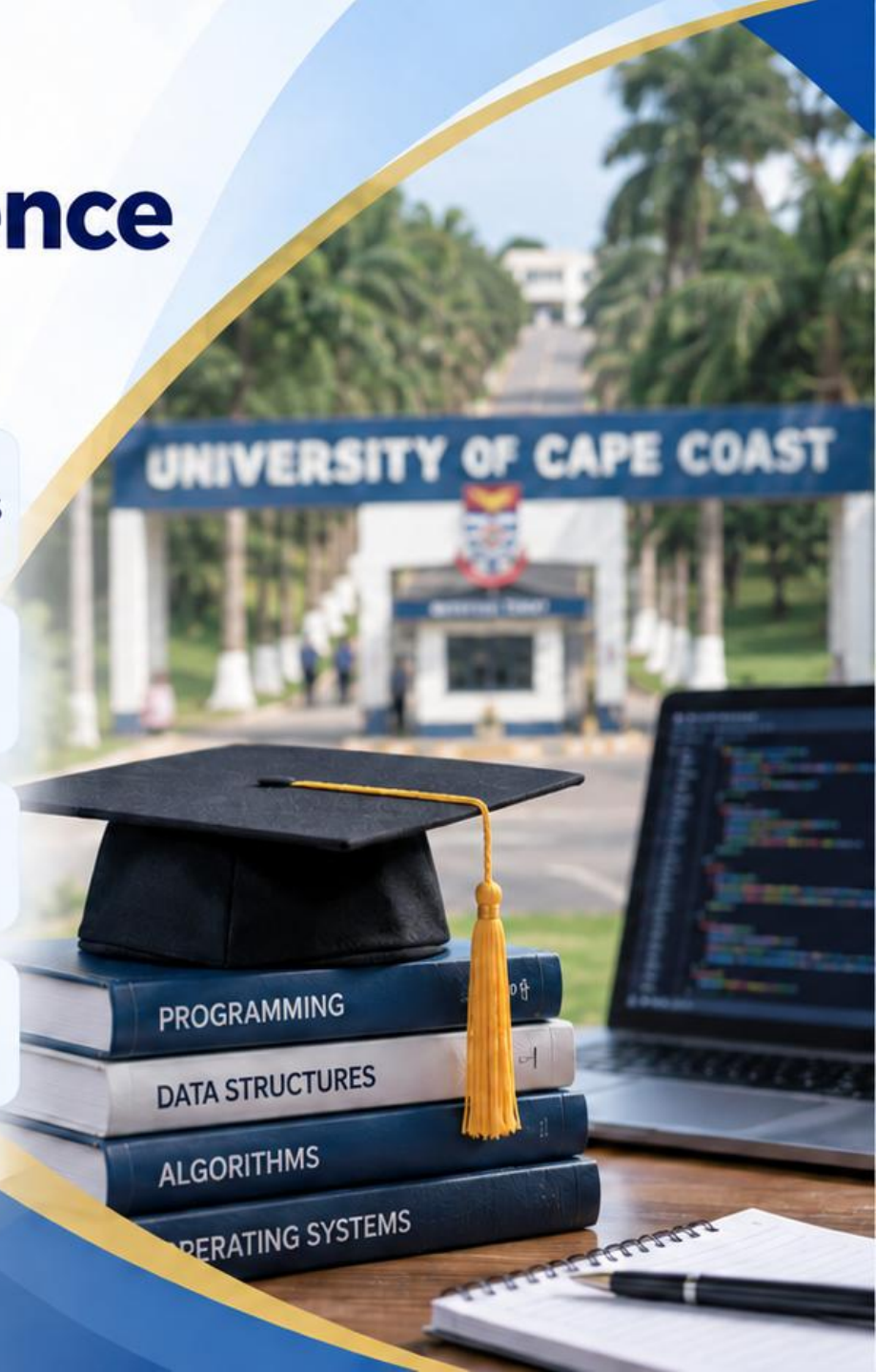
What Careers can you pursue with a BSc in Computer Science?

Career Paths for BSc CS Graduates

- Software Engineering
- Web and Mobile Development
- Artificial Intelligence
- Machine Learning
- Data Science
- Cybersecurity
- Cloud Computing
- DevOps and Cloud-Native Engineering
- Database Engineering
- Computer Networking
- Systems Programming
- Game Development
- Embedded Systems and IoT
- Research and Academia
- Business and Systems Analysis
- IT Consulting
- Technology Entrepreneurship

What Does the Computer Science Degree Give You?

- Programming
- Algorithms
- Data structures
- Computer architecture
- Operating systems
- Databases
- Computer networks
- Software engineering
- Artificial intelligence
- Cybersecurity
- Mathematics
- Statistics
- Theory of computation
- Problem solving



A computer scientist is a builder

> “**design** algorithms, **build** software, **develop** intelligent systems, **analyse** data, **secure** systems and **create** new computing solutions.”

The Competitive Graduate Profile

- BSc Computer Science
- \+ Strong Programming Skills
- \+ Professional Certification
- \+ Digital Badges
- \+ Projects and Portfolio
- \+ Internship/Experience
- \+ Communication and Teamwork
- = Competitive Computer Science Graduate



Certifications vs Digital Badges

Both matter, but they do different things.



Professional certification

- Usually broader, exam-based and recognised by employers or industry bodies.

Examples:
CCNA, Security+, AWS Cloud Practitioner, Azure Fundamentals, ISC2 CC



Digital badge

- Usually smaller and skill-specific: a course, lab, learning path or practical achievement.

Examples:
Microsoft Learn badges, Cisco NetAcad badges, AWS Skill Builder, Google Cloud Skills Boost

Software Engineering

Career roles, certifications and badges that can support this pathway.

Possible roles

- Software Engineer
- Backend Developer
- Frontend Developer
- Full-Stack Developer

Credentials to explore

- GitHub Foundations
- GitHub Actions
- AWS Developer
- Oracle Java

Skills to practise

Python/Java/C++ • Git & GitHub • APIs • Testing • Software Design

Artificial Intelligence

Career roles, certifications and badges that can support this pathway.

Possible roles

- AI Engineer
- Generative AI Engineer
- AI App Developer
- AI Researcher

Credentials to explore

- Azure AI-900
- AWS AI Practitioner
- Microsoft Applied Skills
- Google AI badges

Skills to practise

Programming | Algorithms | Linear algebra | Statistics | Responsible AI

Machine Learning Engineering

Career roles, certifications and badges that can support this pathway.

Possible roles

- ML Engineer
- MLOps Engineer
- Applied Scientist
- Research Engineer

Credentials to explore

- AWS ML Engineer
- Python ML badges
- TensorFlow/ PyTorch
- Cloud ML labs

Skills to practise

Python | NumPy/pandas | Model evaluation | Cloud | MLOps

Data Science

Career roles, certifications and badges that can support this pathway.

Possible roles

- Data Scientist
- Data Analyst
- Research Data Analyst
- BI Analyst

Credentials to explore

- Azure DP-900
- IBM Data Science
- Kaggle certificates
- Google Data

Skills to practise

Python/R • SQL • Statistics • Visualization • Research methods

Data Engineering

Career roles, certifications and badges that can support this pathway.

Possible roles

- Data Engineer
- Big Data Engineer
- Analytics Engineer
- Data Platform Engineer

Credentials to explore

- Azure Data
- AWS Data
- Google Cloud Data
- SQL badges

Skills to practise

SQL • ETL/ELT • Data warehouses • Cloud storage • Distributed systems

Cybersecurity

Career roles, certifications and badges that can support this pathway.



Possible roles

- Cybersecurity Analyst
- Security Engineer
- SOC Analyst
- Penetration Tester

Credentials to explore

- ISC2 CC
- Security+
- Cisco CyberOps
- Microsoft SC-900

Skills to practise

Linux • Networking • Threats • Access control • Ethical practice

Ethical Hacking & Penetration Testing

Career roles, certifications and badges that can support this pathway.

Possible roles

- Ethical Hacker
- Red-Team Specialist
- Vulnerability Analyst
- App Security Tester

Credentials to explore

- Security+
- eJPT
- OSCP later
- Web security labs

Skills to practise

Linux • Web technologies • Cryptography • Authorization • Reporting

Cloud Computing

Career roles, certifications and badges that can support this pathway.

Possible roles

- Cloud Engineer
- Cloud Developer
- Cloud Architect
- Cloud Security Engineer

Credentials to explore

- AWS Cloud Practitioner
- Azure Fundamentals
- Google Cloud Digital Leader
- AWS Developer

Skills to practise

Cloud concepts • Storage • Compute • IAM • Cost awareness

Cloud certification pathways

Start with fundamentals, then move into role-based credentials.

AWS

AMAZON WEB SERVICES

1 → 2 → 3

Cloud Practitioner → Developer or Solutions Architect Associate → specialist/professional growth

Microsoft Azure

CLOUD FOR A BRIGHTER TOMORROW

1 → 2 → 3

Azure Fundamentals → AI/Data/Security Fundamentals → role-based Azure credentials

Google Cloud

BUILD WHAT'S NEXT

1 → 2 → 3

Cloud Digital Leader → Associate Cloud Engineer → professional-level credentials

DevOps & Platform Engineering

Career roles, certifications and badges that can support this pathway.

Possible roles

- DevOps Engineer
- Platform Engineer
- SRE
- Build & Release Engineer

Credentials to explore

- LFCA/LFCS
- KCNA
- CKA
- CKAD

Skills to practise

- Linux
- Git
- Docker
- Kubernetes
- CI/CD

Linux & Systems Programming

Career roles, certifications and badges that can support this pathway.

Possible roles

- Linux Engineer
- Systems Programmer
- Infrastructure Engineer
- Embedded Software Engineer

Credentials to explore

- LFCA
- LFCS
- Linux+
- Red Hat

Skills to practise

- Linux
- C/C++
- Operating systems
- Bash
- Virtualization

Database Engineering

Career roles, certifications and badges that can support this pathway.

Possible roles

- Database Developer
- DB Administrator
- Database Engineer
- Data Architect

Credentials to explore

- SQL badges
- Oracle
- Microsoft
- Cloud databases

Skills to practise

SQL | Relational design | Normalization | Transactions | NoSQL

Networking

Career roles, certifications and badges that can support this pathway.

Possible roles

- Network Engineer
- Network Software Engineer
- Cloud Network Engineer
- Network Security Engineer

Credentials to explore

- Cisco CCNA
- Cisco NetAcad
- Network+
- Wireless basics

Skills to practise

- IP addressing
- Routing
- Switching
- Distributed systems
- IoT

Embedded Systems & IoT

Career roles, certifications and badges that can support this patthway.

Possible roles

- Embedded Software Engineer
- Firmware Developer
- IoT Engineer
- Robotics Software Engineer

Credentials to explore

- C/C++ badges
- IoT badges
- Linux basics
- Hardware labs

Skills to practise

C/C++ | Microcontrollers | Sensors | Real-time computing | Networking

Game Development & Computer Graphics

Career roles, certifications and badges that can support this pathway.

Possible roles

- Game Developer
- Graphics Programmer
- Simulation Developer
- AR/VR Developer

Credentials to explore

- Unity badges
- Unreal learning
- C++/C#
- Graphics courses

Skills to practise

Linear algebra • Graphics • C++/C# • Game engines • Portfolio

Research & Academia

Computer Science is also a research discipline.



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Build projects around your career path

Projects convert learning into evidence.



Software Engineering

Web application • mobile app • API • database system



AI / ML

Classifier • recommendation system • NLP app • computer vision app



Cybersecurity

Secure application • home security lab • log-analysis project



Data Science

Dashboard • predictive model • data-analysis notebook



Cloud

Cloud-hosted app • serverless app • containerized application

GitHub should become part of your professional identity

Begin using GitHub early and let it show your growth.

1



Source code

Store coursework and personal projects with clear folder structure.

2



Version control

Practise commits, branches and collaboration.

3



Documentation

Use README files to explain what the project does.

4



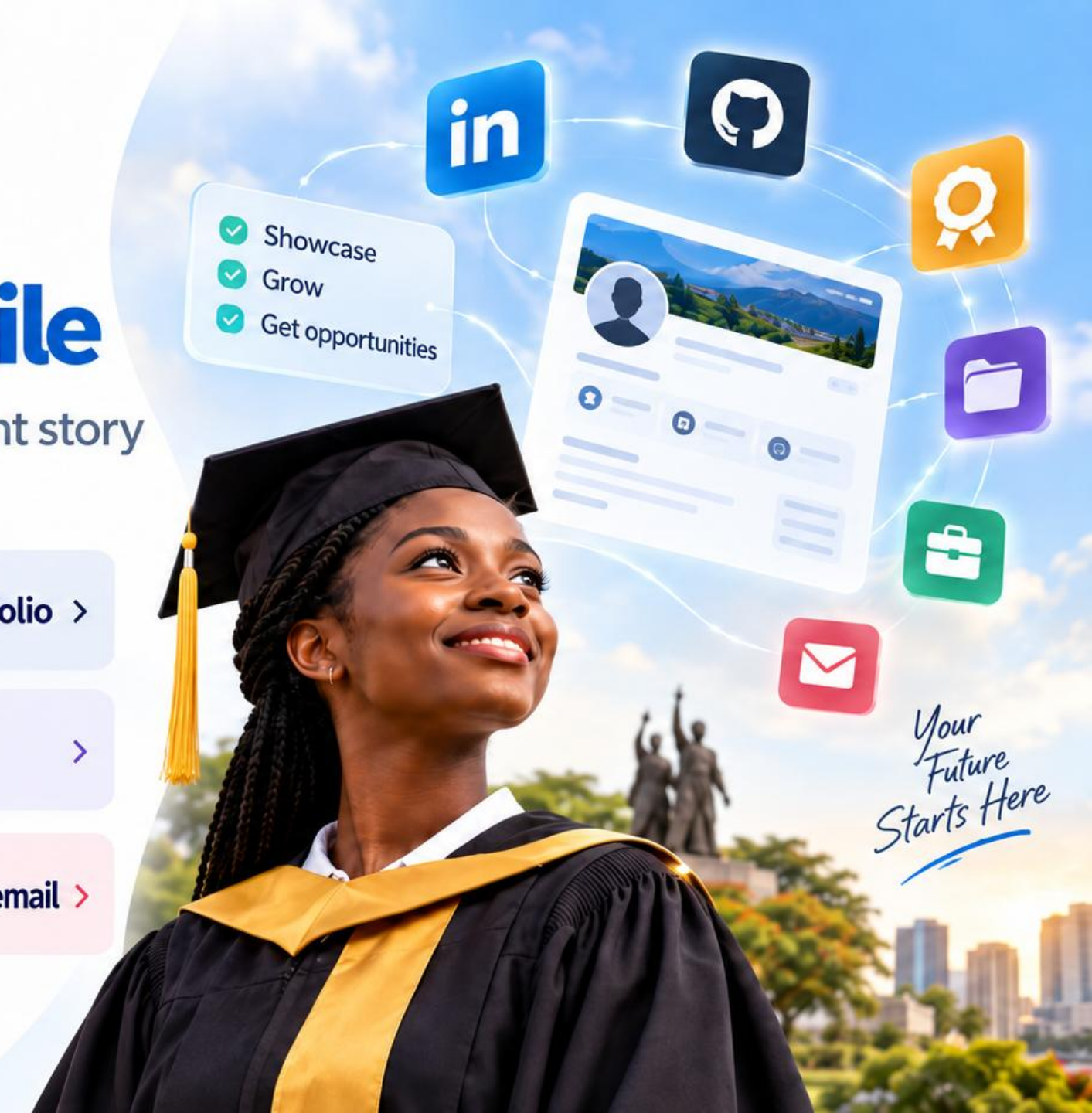
Progress over time

Level 100 → Level 200 → Level 300 → Final-year project.

Build a digital professional profile

Your online profile should tell a coherent story before you graduate.

- LinkedIn profile
- GitHub portfolio
- Digital badges
- Projects
- Internships
- Professional email



Four-year roadmap

Grow from freshman curiosity to graduate readiness.

1 Level 100



Explore

- Intro to computing and core concepts
- Earn badges
- Basic coding (e.g. Python)
- Linux and essential tools
- Build small projects

2 Level 200



Specialise

- Choose one or two directions (e.g. software, data, cybersecurity, cloud)
- Take core and elective courses
- Get your first certification
- Work on intermediate projects

3 Level 300



Build

- Develop real-world projects
- Join labs, research or internships
- Participate in competitions
- Strengthen professional skills (GitHub, teamwork, communication)

4 Level 400



Position

- Complete a portfolio
- Work on a final-year project
- Prepare for the job market or graduate studies
- Obtain industry-recognised certifications
- Be job-ready

Skills no certification can replace

- Algorithms
- Data structures
- Programming
- Mathematics
- Debugging
- Logical reasoning
- Communication
- Teamwork
- Technical writing
- Research ability
- Professional ethics

The role of AI in your career

Use AI as a learning assistant, not as a replacement for thinking.



Use AI for support

Code explanation, debugging assistance, documentation, brainstorming, testing and prototyping.



Still understand the work

The code, algorithm, security implications, accuracy, bias, testing and ethics.



Strong message

Do not allow AI to replace your thinking while you are learning how to think computationally.

Graduate with more than a transcript

“Your degree gives the foundation. Your projects prove what you can build. Your certifications validate specialisation. Your attitude determines how far you will go.”

Final question for you

When you graduate four years from now, what evidence apart from your degree will prove you are ready?

1



Which CS career interests me?

Explore roles and find what excites you.

2



What skills does that career require?

Understand the technical and soft skills you need.

3



Which courses in my BSc CS support it?

Identify key modules that build relevant skills.

4



Which certification aligns with it?

Find industry-recognised certifications to strengthen your profile.

5



Which badges can I earn along the way?

Collect micro-credentials and achievements to show your progress.

6



What projects can I build to demonstrate the skill?

Create real-world projects that showcase your abilities.

Your future is built by what you do next.

Online Courses

University-quality / full learning paths (free)

MIT OpenCourseWare (OCW)
full course materials (notes, exams, videos) for many CS/IT topics.

CS50 (Harvard)
top-tier intro CS + programming; you can start free (audit)

freeCodeCamp
100% free coding curriculum + projects + certifications.

edX (audit mode)
many courses can be audited for $0 (cert usually paid).

Online Courses

Official vendor training (free learning content)

Microsoft Learn
free learning paths for Azure, security, Microsoft 365, etc.

AWS Skill Builder / AWS Digital Training
free AWS courses (and more via Skill Builder).

Cisco Networking Academy (NetAcad)
free self-paced courses (networking, cybersecurity, Linux, Python).

Google Career Certificates (on Coursera)
popular IT Support / Cybersecurity / Data Analytics tracks (often trial/aid options).

Online Courses



Cybersecurity hands-on (free labs/training)

- **PortSwigger Web Security Academy** – excellent free web app security lessons + labs.
- **TryHackMe** – browser-based security practice; many beginner resources/rooms are free (some paths require paid).



Data & analytics (free, practical)

- **Kaggle Learn** – short, hands-on micro-courses (Python, Pandas, SQL, ML).



Git / GitHub & collaboration (free)

- **GitHub Skills** – interactive beginner-to-intermediate mini-courses (GitHub basics, Markdown, etc.).

Big Course Platforms

Coursera
University + industry courses (often audit/free trial).

edX
University courses (often free to audit).

Udemy
Huge catalog; usually paid (frequent discounts).

Pluralsight
Strong for IT/professional tech (subscription; trials sometimes).

LinkedIn Learning
Practical career courses (subscription; trials sometimes).

Skillshare
Creative + tech basics (subscription).

Free, beginner-friendly learning

Khan Academy

Khan Academy

Basics in computing, math foundations.

freeCodeCamp

freeCodeCamp

Free web dev + coding paths with projects.

THE ODIN PROJECT

The Odin Project

Free full-stack web development.

w3schools

W3Schools

Quick tutorials + references (web tech, SQL).

GeeksforGeeks

GeeksforGeeks

Coding + CS/IT topics, interview prep.

tutorialspoint

TutorialsPoint

Wide IT tutorials (quick reference style).

CSC101: Introduction to Computing (3 Credit Hour)

Course Description

- Explores essential concepts in computer systems, hardware, software, data representation, and operating systems.
- Covers the basics of computer networks, the Internet, and cloud computing architectures.
- Introduces computational thinking, algorithms, and foundational programming concepts to solve real-world problems.
- Examines databases, information systems, cybersecurity principles, artificial intelligence, and emerging technologies.
- Analyzes the ethical and societal impacts of technology while building critical skills needed for advanced CS coursework.

Course Structure

Week	Content
1	Introduction to Computing
2	Computer Hardware and Basic Architecture
3	Data Representation and Digital Information
4	Computer Software and Operating Systems
5	Computer Networks, Internet and Cloud Computing
6	Computational Thinking and Algorithms
7	Introduction to Programming Concepts
8	Databases, Data and Information Systems
9	Cybersecurity, Ethics and Responsible Computing
10	Emerging Technologies and the Future of Computing

Assessment

End-of-Semester Examination
60%

Continuous Assessment —
40%

Assessment Component	Weight
Quizzes / Class Exercises	20%
Individual Assignment	10%
Group Assignment / Presentation	10%
Total Continuous Assessment	40%

Reading List

1. Vermaat, M. E., Sebok, S. L., Freund, S. M., Campbell, J. T., & Frydenberg, M. (2023). *Discovering computers: Digital technology, data, and devices*. Cengage Learning.

2. Evans, A., Martin, K., & Poatsy, M. A. (2022). *Technology in action complete* (17th ed.). Pearson.

3. Forouzan, B. A. (2021). *Foundations of computer science* (4th ed.). Cengage Learning.

4. Morley, D., & Parker, C. S. (2021). *Understanding computers: Today and tomorrow, comprehensive* (17th ed.). Cengage Learning.

5. Reynolds, G. W. (2022). *Ethics in information technology* (7th ed.). Cengage Learning.

6. Stallings, W. (2022). *Computer organization and architecture: Designing for performance* (11th ed.). Pearson.

7. Russell, S. J., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.